

CHAPTER 2

GROWN, NOT CRAFTED

Scene: A man and a woman are sitting in a restaurant in daytime. The woman's voice is low and intense.

WOMAN: And the thing is, I just can't seem to have a reasonable conversation about this with anyone. Everyone in his family except for him is a terrible person, and he says he has to work hard at not being terrible himself, and sometimes he *is* terrible, including with me. On my own side, my family struggles with depression. And my parents are after me to have a baby, and my friends have a zillion opinions that seem based on nothing but their own experiences with their own partners and kids. Should I be looking for a sperm donor? Should I just give up on having children? And if I do have a baby, will it grow up happy and kind?

MAN: You've come to the right place! Nobody can predict baby outcomes better than I can! I know all about how babies are made!

WOMAN: You... what? I don't need to know how babies are made! I need to know how my baby will turn out!

MAN: Well, if you know how babies are *made*... then what else could you possibly need to know?

WOMAN: My first thought was, "my baby's actual genetics," except—

MAN: Ah! Then I have the solution! Individual whole-genome sequencing is quite affordable these days. Just make an embryo with one of your eggs and your husband's sperm, and have the genome sequenced before deciding whether to implant it. That gene sequence will tell you everything there is to know about your baby.

WOMAN: But do we know enough about genes that are associated with things like “being a terrible human being” or “being happy”? Enough to tell me—

MAN: Once you know all the DNA bases in your baby’s genes, and you can look them up at any time, your baby’s genetics will be transparent to you. There will exist no truth about its genes that you don’t know.

WOMAN: I’d get a file containing three billion inscrutable letters like “CATTCA.” It would take me a hundred years to read them all. I would learn nothing even if I tried. Even if those billions of inscrutable letters do have a huge influence on my baby’s fate, raw DNA letters don’t really tell me how my child’s brain works, what thoughts will happen inside that brain after my child grows up...

MAN: Oh, you mean you don’t know about physics! But that’s straightforward! Once you know how protons and neutrons and electrons interact, you’ll know everything that exists to be known about brains, because you’ll understand every event that goes on inside the neurons.

WOMAN: Let’s change the subject.

BEFORE WE CAN EXPLAIN WHY ARTIFICIAL SUPERINTELLIGENCE achieved using anything like modern methods would inevitably go wrong, we need to quickly survey those modern methods: how they work, what they produce, and what AI engineers have in common with a mother who knows only her baby’s DNA.

The most fundamental fact about current AIs is that they are *grown*, *not crafted*. It is not like how other software gets made—indeed it is closer to how a human gets made, at least in some important ways. Namely, engineers understand the process that results in an AI, but do not much understand what goes on inside the AI minds they manage to create.

Suppose an engineer, with no deep understanding of language or grand theory of intelligence, would like to make a machine that talks sensibly. How might they go about it?

They want to write some text and have a machine continue writing that text in a sensible manner.

They decide to start by teaching it that, in a sentence that begins once

upon a ti, the next letter is probably “m.”

They *could* create a program that just checks whether the sentence fragment is Once upon a ti, and then produces the letter m. But that wouldn’t work on other sentence fragments like Four score and sev, and they want the machine to be good at completing all sorts of sentences, or at generating conversations or essays. They even want it to sensibly complete sentences that no person has ever typed before.

If they use modern AI methods, what they do next goes, approximately, something like this:

1. First, they take the sentence fragment—Once upon a ti, in our example—and render it as a series of numbers by, say, associating A with 1, B with 2, C with 3, and so on, plus some numbers for spaces and punctuation. Once upon a ti becomes a series of numbers like 15 14 3... This is called the **input**.
2. Next, they acquire a computer that can store loads and loads of numbers. Each slot for a number is called a **parameter**. (As of early 2025, cutting-edge AIs use a few trillion parameters.)
3. They fill that storage with numbers; to oversimplify, let’s say those numbers are randomly selected. The numbers in the slots are called **weights**.
4. Then, they determine the **architecture**: the rules for how to combine their input (like the Once upon a ti sequence of 15 14 3...) with the weights in the parameters. Something like, “I’ll multiply each input-number with the weight in the first parameter, and then add it to the weight in the second parameter, and then I’ll replace it with zero if it’s negative, and then...” They pick a lot of operations like that—hundreds of billions, linking every single weight into the calculation.
5. All these operations spit out a set of **output** numbers, which they interpret as a prediction about what letter comes next. In our example, they’d treat the first output number as a probability of A, and the second output number as a probability of B, and so on.
6. Next, they “train” their budding machine intelligence using a

process called **gradient descent**.

Since the initial weights are random, when they first run this program, it spits out nonsense. Maybe it'll say that the letter b is 65 percent likely to come next, and that the letter m is only 1 percent likely to come next.

But here's the thing: If they chose the architecture just right, they can calculate the role that every single parameter played in determining the final result of all that arithmetic.

So now they take every single weight—hundreds of billions of weights—and ask for each one: “If I'd made this number a tiny bit larger or smaller, how much more or less probability would've been given to m, at the end of all that arithmetic?”

This is called the **gradient** for that parameter.ⁱ The gradient says how—and how much—to change the weight in that parameter in order to make the final answer a little more correct.

So then they go ahead and tweak each and every weight according to its gradient. They push every single weight in the direction that makes the answer slightly more correct. Not by hand; they write a program to do it. AI engineers rarely look at any of the numbers; it would take more than a human lifetime to look at them all.

Computer scientists think of this as “descending” toward a “less bad” answer, hence, “gradient descent.”

Doing this *once* will not give a perfect answer, only a slightly less bad answer. But because this entire process can be automated, it can be repeated over trillions of words, called the **training data**, in just a few months, for just a few hundred million dollars, on the world's most advanced computers. (Hopefully our protagonist is wealthy, or works for a big company.) This process is called **training**.

7. Once the machine is all trained up, they can turn the machine's outputs—the probabilities that it generates—into ordinary text that a user sees. If the AI most strongly predicts m as the continuation of Once upon a ti, they add that m on to get Once upon a tim. Then they feed the new extended sequence back in

again to ask for a prediction of the next letter, and get e. They keep going, and the machine starts to talk.

That set of hundreds of billions of weights, tweaked over and over via gradient descent until their most likely predictions look like real human language, is called a large language model (LLM). A “base model,” in particular.

If they want to turn their base model into a helpful LLM, like ChatGPT, there’s one more step: another round of gradient descent on inputs formatted like:

User: What is the capital of Spain?

Assistant: Madrid.

The purpose of this part isn’t to teach the LLM that the capital of Spain is Madrid; the LLM already knows that after being trained on much of the internet. Rather, the idea is to tune the LLM to fill in the text after “Assistant:” with a *helpful answer* rather than a response like “Why the heck are you asking me? Google it yourself,” no matter how common the latter might be in the actual human conversations it was trained on. If our engineer was working for a major corporation that supplied all those computers, this is also the phase where they’d train the AI against swearing and talking about how to hotwire a car, using human (or, lately, AI-generated) ratings about which sorts of answers are most corporately palatable.

And that’s where babies come from, metaphorically speaking.

When it comes to making *actual* babies, would-be parents don’t need to know much science. In this particular case, the parallel holds. AI engineers seeking to make an AI need to know somewhat more than human parents do—but not as much as you might think.

To recap, here’s what today’s AI “parents” do: Engineers choose the AI’s